

Surya Sys.

(AI-Powered Solar Analysis)

Powering India's Solar Revolution through Artificial Intelligence

Presenters:



The Fuel Supply Crisis

Iran & Israel War 2026

The Problem

India relies heavily on natural gas and oil through the Strait of Hormuz. The Iran & Israel War 2026 disrupted this route, cutting nearly half of India's gas imports.

The Result

Yelahanka Power Plant and other major facilities forced to shut down, creating massive electricity supply gaps.

Possible Solution

Electrify India completely using renewable energy, so we won't have to rely on fossil fuels.



Electrification Challenges

A logical solution is to electrify the nation using renewable energy, but new problems emerge:

Insufficient Electrical Sources

India lacks enough renewable capacity to rapidly replace fossil fuels.

Uneven Solar Distribution

Solar energy varies significantly across different regions.

Limited Monitoring

No intelligent forecasting of solar infrastructure deployment.



Regional Transmission Gaps

Rajasthan and Gujarat generate most solar power

Lack of high-capacity infrastructure in solar plants for energy transportation

Northern, Eastern and Southern India face shortages

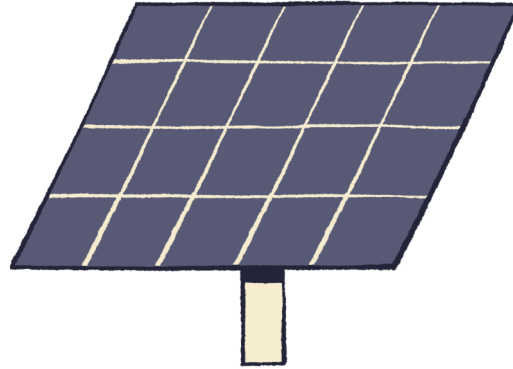
Production Regions

Transmission Gap

Industrial Demand

Solar power becomes geographically trapped in production regions while industrial areas face power shortages and blackouts.

Solar Forecasting Challenges



The Problems

- India receives abundant sunlight, but solar power prediction remains unreliable due to tropical climate and rapidly changing weather.
- HVDC (High Voltage Direct Current) transmission lines and Solar Energy Storage systems exist, but they are very expensive.

Possible Solution

Buy high-infrastructure transmission systems, make solar energy scalable for bad weather and accelerate production during favorable conditions.



Grid Stability Challenges

Traditional power plants provide inertia through large rotating turbines. Solar panels have no moving mechanical parts and cannot provide physical grid stability.

01

Loss of Inertia

No rotating mass to absorb sudden demand changes

02

Frequency Instability

Short drops in solar output cause fluctuations

03

Large-Scale Blackouts

Fragile grid leads to cascading failures

The ONLY Viable Solution

Accelerate solar energy adoption by equalizing distribution through regional solar energy analysis.



Project Vision



Automate Analysis

High-resolution satellite imagery for solar potential evaluation



Transparent Subsidies

Instant calculations for homeowners based on government policies



Centralized Hub

Manage solar installation history and data



Expert Support

Real-time AI and human assistance

An illustration of a modern house with a large array of solar panels installed on its roof. Two people are standing on the roof, one pointing towards the panels. The house is surrounded by lush green trees and a clear blue sky with scattered white clouds. In the background, rolling hills and a distant cityscape are visible under a bright sun.

How It Works

An AI-powered solar analytics dashboard evaluates rooftop solar potential instantly using satellite imagery, machine learning, and interactive visualization.



Location Selection

Users select home location on digital map



AI Detection

System detects rooftop area and calculates usable space



Capacity Estimate

Estimates solar system capacity in kilowatts



Subsidy Calculation

Provides instant subsidy information

Technical Implementation

Frontend Stack

- React 19 with TypeScript
- Vite 7 build tool
- Tailwind CSS v4
- Framer Motion animations
- TanStack Query

Backend & ML

- FlaskPython REST API
- PostgreSQL with SQLAlchemy
- RetinaNet (PyTorch) via Roboflow
- ArcGIS JavaScript API
- Ably Realtime SDK



Satellite Integration

ArcGIS for precision mapping



ML Detection

Roboflowmodelsfor panel
detection



3D Visualization

Real-time SceneView rendering



Key Benefits

Accelerated Adoption

Automates rooftop solar potential analysis, enabling faster renewable energy adoption

Transparent Subsidies

Provides clear, instant subsidy information for solar installations

Scalable Platform

Supports millions of households and Government based schemes

Support Ecosystem: AI chatbot for scheme queries, live expert chat via Ably Realtime SDK, and feedback system via Formsfree.